JOHN LOUIS L. ALANO

Mechanical Design & Product Development Engineer

Portfolio · johnlouislalano@gmail.com
+63 924 100 9727 · Manila, Philippines
linkedin.com/in/johnlouislalano

PROFESSIONAL SUMMARY

Mechanical Design Engineer with experience in mechanical systems design, product engineering, engineering analysis, prototyping, and multidisciplinary vehicle development. Mechanical Lead and Vehicle Designer for the first Philippine team and sole ASEAN and Asian representative in the Shell Eco-marathon Autonomous Driving Competition in Poland, finishing 8th Overall internationally. Experienced in 3D CAD modeling, FEA, CFD, DFMA, GD&T, engineering drawings, mechanical system integration, and prototype fabrication. Currently leads product development projects, applying practical design and manufacturing principles to transform concepts into functional and manufacturable systems.

TECHNICAL SKILLS

Mechanical Design	3D CAD Modeling, Mechanical Systems Design, Assembly Design, Chassis Design, Mechanism Design, Design Optimization, Engineering Calculations
CAD & Engineering Software	Solid Edge, SolidWorks, Autodesk Inventor, MATLAB/Simulink, ANSYS, SimScale, KeyShot
Engineering Analysis	FEA, CFD, Static Structural Analysis, Modal Analysis, Stress Analysis, Fluid-Flow Analysis
Manufacturing & Prototyping	DFMA, GD&T, Engineering Drawings, Tolerance Analysis, DfAM, 3D Printing, CNC Machining, Manual Fabrication, Rapid Prototyping
Electronics & Embedded	Basic Electronics, ESP32, Arduino IDE, C++, Sensor Integration, Motor Control

WORK EXPERIENCE

Yuro Product Development and Design Solutions — Oct 2024 – Present

Manila, Philippines | Mechanical Design Consultant (2026 – Present); Product Development Lead (Oct 2024 – 2025)

- Led mechanical development from concept and CAD through engineering analysis, prototyping, fabrication, and validation.
- Designed mechanical components and assemblies using Solid Edge, applying DFMA, GD&T, engineering calculations, and manufacturability considerations.
- Provided technical consultation on mechanical architecture, prototyping, design optimization, and multidisciplinary system integration.

DBBC Engineering Consultancy — Jul 2025 – Sept 2025

Quezon City, Philippines | Mechanical Design Intern

- Performed HVAC, fluid-flow, and hydraulic analyses for equipment selection and system design using Revit and ANSYS.
- Conducted field validation and commissioning, comparing calculated and measured system performance.

PROJECTS AND LEADERSHIP

Mechanical Lead – Shell Eco-marathon Autonomous Driving Programme — Nov 2025 – Jun 2026

Team Wired, Polytechnic University of the Philippines | International competition, Poland

- Led mechanical development of an autonomous prototype vehicle that finished 8th Overall in Shell Eco-marathon Poland 2026.
- Directed vehicle architecture, structural design, CAD, FEA, CFD, fabrication, assembly, and mechanical subsystem integration.
- Designed and optimized the vehicle's modularity achieving 57.7% reduction in volume for overseas deployment.
- Coordinated mechanical integration with electrical, controls, embedded, and autonomous systems within a tight five-month development timeline.

Mechanical Head & Technical Instructor – PUP WIRED — Sep 2024 – Jul 2026

- Led mechanical R&D and innovation initiatives, collaborating with cross-functional teams on engineering projects and competitions.
- Trained 100+ students in Solid Edge, covering 3D CAD, assemblies, engineering drawings, simulation, and mechanical analysis.

Head of Product Development – PUP Hygears — Mar 2025 – Jul 2025

Polytechnic University of the Philippines

- Led the organization's design-to-product workflow, overseeing CAD development, DFMA, prototyping, testing, and design validation.
- Directed multidisciplinary teams in developing functional and manufacturable engineering prototypes.

Autonomous Electromagnetic Variable Tuned Mass Damper — Undergraduate Thesis

Polytechnic University of the Philippines

- Designed and simulated an electromagnetic variable-tuned mass damper for adaptive vibration suppression and energy harvesting using dynamic analysis, MATLAB/Simulink, and FEA.
- Experimentally validated vibration suppression, achieving up to 79.72% acceleration reduction and shifting dominant vibration frequencies from 67–79 Hz to lower bands near 29.7 Hz.
- Developed an ESP32-based control system using Arduino IDE, integrating sensors and actuation for automated tuning based on vibration conditions.

ACHIEVEMENTS

- 8th Overall, Shell Eco-marathon Autonomous Driving Competition — Poland 2026
- 7th Academic Research Conference for Mechanical Engineers - 3rd Place
- 6th Midea International HVAC Design Contest - Participant
- Autodesk Fusion 360 Competition JPSME National Students' Conference - Participant

CERTIFICATIONS

- Solid Edge Associate | Sep 2024
- CSWA (Certified SOLIDWORKS Associate) | Sep 2025
- Lean Six Sigma White Belt — Six Sigma PH | Jul 2026

EDUCATION

Bachelor of Science in Mechanical Engineering — Oct 2022 – Sept 2026 (Expected)

Polytechnic University of the Philippines | Manila